

Simulations of Interacting Active Brownian Particles in Curved Geometries

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What are we doing here?

- Active matter is a thriving field of modern physics research.
- Active matter = nonequilibrium soft matter + self-propulsion.
- Often used to model the behavior of biological systems.
- Active Brownian Particles (ABPs) are individual agents move on a surface with an intrinsic velocity.
- We extend the basic model with Vicsek-style orientational alignment and curved surface geometries.
- The goal: determine if/how curved surfaces effect the behavior of the model.

Basic Active Brownian Particles

The simplest mathematical model of ABPs is given below:

$$\begin{aligned}\mathbf{r}_i(t + dt) &= \mathbf{r}_i(t) + v_0 \hat{\mathbf{n}}_i dt \\ \theta_i(t + dt) &= \theta_i(t) \\ &\quad + \sqrt{2Ddt} \xi_i(t)\end{aligned}$$

$$i = \{1, 2, \dots, N\}$$

$\mathbf{r}_i$: Vector position of particle i . In Euclidean space, $\mathbf{r}_i = \langle x_i, y_i \rangle$

θ_i : Orientation of particle i .

$\hat{\mathbf{n}}_i$: Unit heading vector of particle i . In Euclidean, $\hat{\mathbf{n}}_i = \langle \cos \theta_i, \sin \theta_i \rangle$.

$\xi_i(t)$: Sample from standard normal distribution.

v_0 : Intrinsic self-propulsion velocity of all particles.

D : Diffusion coefficient, controls “how random” the motion is.

N : Number of particles

Vicsek Orientational Alignment

- In the Vicsek model, particles orient themselves in the average direction of their neighbors.
- Over time, and with a sufficiently large interaction radius, particles “flock” together.
- Modelled as an additional term in the time-step for θ :

$$\theta_{Vicsek,i}(t + dt) = C \sin(\theta_{mean} - \theta_i)dt$$

$$\theta_{mean} = \text{atan2} \left(\sum_{j \neq i} \sin \theta_j, \sum_{j \neq i} \cos \theta_j \right)$$

- C is the *coupling strength* parameter.

Differential Geometry for Dummies

- Curved surfaces may be generalized with the concept of *manifolds*.
- General manifolds are complicated objects, but we are concerned with smooth (differentiable) two-dimensional Riemannian manifolds (embedded in $\mathbb{R}^3$).
- Every point on a differentiable manifold has a local tangent space, allowing us to define velocity vectors (our θ_i are angles measured on the tangent space).
- Riemannian manifolds are equipped with extra structure known as a *metric* g , which allows the measurement of distances.
- The metric is defined with two indices: g_{ij} , and for two-dimensional manifolds is given as a 2×2 tensor.
- A manifold may be given a coordinate mapping, which in general in two-dimensions we'll call q_1 and q_2 .

Differential Geometry the Lightly Educated

- Tangent spaces rotate due to curvature across the surface.
- Without correcting for this, our particles experience a “phantom torque” following the surface curvature.
- We add another term to θ , called the *spin connection*.

$$\omega_{\mu}^{ab} = e_{\nu}^a \Gamma_{\sigma\mu}^{\nu} e^{\sigma b} - e^{\nu b} \partial_{\mu} e_{\nu}^{\alpha}$$

$$\theta_{\text{spin},i}(t + dt) = \omega_{\mu}^{ab} dq^{\mu}$$

- e_{ν}^a are *vierbein* defined implicitly w.r.t. the *metric* through $g_{ij} = e_i^a \delta_{ab} e_j^b$. They allow us to translate between the coordinates on the manifold and the local tangent space.
- Γ_{jk}^i are the *Christoffel symbols*, also defined with respect to the metric, $\Gamma_{jk}^i = \frac{1}{2} g^{ii} (\partial_j g_{ik} + \partial_k g_{ij} - \partial_i g_{jk})$. They measure misalignment between straight-line vs curved-surface motion.

Further Consequences of Differential Geometry

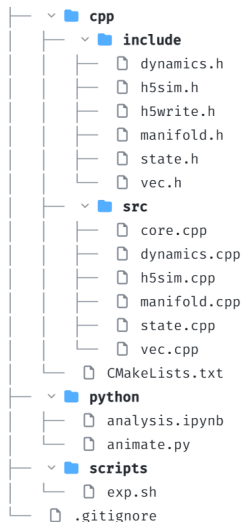
- Our Vicsek update involves checking which other particles are nearby, within some interaction radius.
- The Euclidean distance, or the Pythagorean theorem, $d(\mathbf{a}, \mathbf{b}) = \sqrt{dq_1^2 + dq_2^2}$ (where $dq_1 \approx q_{1,a} - q_{1,b}$ and $dq_2 \approx q_{2,a} - q_{2,b}$) doesn't accurately measure distance on curved manifold.
- We need the geodesic distance, which with a diagonal metric is approximately:

$$d(\mathbf{a}, \mathbf{b}) = \sqrt{g_{11}dq_1^2 + g_{22}dq_2^2}$$

- We also can't directly average angles in the Vicsek angles, we first need to *parallel transport* the tangent spaces at the two points so the angles mean the same thing.

Simulation Setup

- The core of the majority of our simulation is built in C++
- C++ is generally faster than Python, particularly for many nested loops of numerics.
- We make use of modular header files to keep code organized.
- Data is analyzed and visualized in Python, since ecosystems are both more robust and easier to work with there.
- Parameter sweeps over D and C are performed in parallel with a bash script calling the compiled C++ executable.



Specific Manifolds

- I chose to implement a spherical and toroidal manifold, as well as regular Euclidean space as a control.
- The coordinate chart on the sphere is as usual:
 $q_1 = \phi \in [0, 2\pi)$ (azimuthal angle), $q_2 = \alpha \in [0, \pi)$ (polar angle, usually θ but that's taken). For radius R , the metric is:

$$g_{\text{sphere}} = \begin{bmatrix} R^2 \sin^2 \alpha & 0 \\ 0 & R^2 \end{bmatrix}$$

- The coordinate chart on the sphere is $q_1 = \phi \in [0, 2\pi)$ (toroidal angle) $q_2 = \psi \in [0, 2\pi)$ (poloidal angle). With major radius R and minor radius r , the metric is:

$$g_{\text{torus}} = \begin{bmatrix} (R + r \cos \psi)^2 & 0 \\ 0 & r^2 \end{bmatrix}$$

The Vaguest Possible Pseudocode Overview

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1: procedure STEPSIMULATION( $\Delta t$ )
2:   Phase 1: Geometry & Dynamics
3:   for each particle  $i$  do
4:      $\theta_i \leftarrow \theta_i + \omega + \sqrt{2D\Delta t} \cdot \mathcal{N}(0, 1)$   ▷ Update Orientation
5:      $q_i \leftarrow q_i + \text{Vielbein}(q_i) \cdot \vec{v}(\theta_i)\Delta t$   ▷ Update Position
6:      $\text{Manifold.wrap}(q_i)$   ▷ Boundary conditions
7:   end for
8:   Phase 2: Interaction & Alignment
9:   for each pair  $(i, j)$  within  $R_{int}$  do
10:    if  $d(i, j) < 2s_{radius}$  then
11:       $\text{ResolveCollision}(i, j)$   ▷ Elastic reflection
12:    end if
13:     $\text{ParallelTransport}(\theta_i \leftrightarrow \theta_j)$   ▷ Align tangent spaces
14:     $\theta_i \leftarrow \theta_i + \sum \sin(\theta_{\text{mean}} - \theta_i)\Delta t$   ▷ Vicsek alignment
15:  end for
16: end procedure

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Investigation of the Scientific Question

- At the beginning, I said the goal was to see how different manifolds exhibit different behavior.
- The specific behavior we're looking at is the phase-transition that comes from the interplay of the diffusion and coupling coefficients.
- At high diffusion, the particles act like a gas, moving randomly and overcoming their orientational alignment.
- At high coupling, the particles act like a fluid, flowing in groups along the surface.
- Each “experiment” run consists of running the algorithm on the previous slide for a large number of timesteps on each of our three manifolds, with four different values of C and D .
- By computing the global order parameter for each manifold over different sets of parameters, we can see how alignment behaves as a function of C , D , and the manifold.
- For fun, we also animate the particle trajectories.

Global Order Parameter

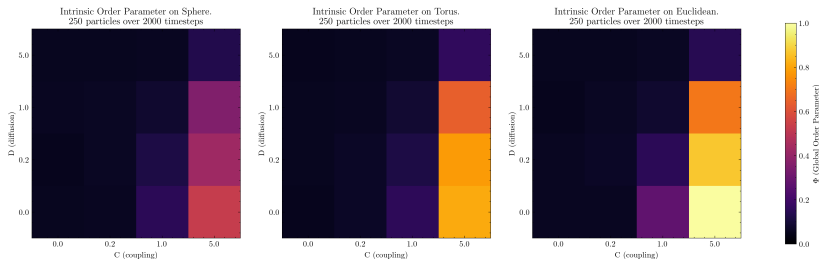
- To compare the alignment behavior across manifolds, we modify the typical Vicsek global order parameter:

$$\Phi = \left\langle \left| \frac{1}{N} \sum_i \mathbf{v}_i \right| \right\rangle$$

- It ranges from 0 to 1, with 0 being complete random alignment (gas-like behavior) and 1 being complete global alignment (flocking) of all particles.
- $\langle \cdot \rangle$ is a time average over the end of the simulation, ensuring steady-state behavior rather than a transient coincidence.
- Particles moving in the same direction on a curved surface will not have the same $\mathbf{v}_i$ vector in 3D space; we must first parallel transport the vectors to the same location on the surface, with $\bar{\theta}_i$ equal to the orientation after transport.

$$\Phi_{\text{intrinsic}} = \left\langle \left| \frac{1}{N} \sum_i e^{i\bar{\theta}_i} \right| \right\rangle$$

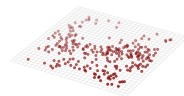
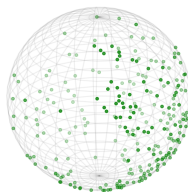
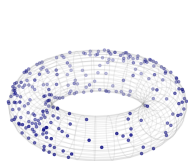
Global Order Parameter Results



- In general, we see what we expect: particles align far more in low diffusion, high coupling regimes.
- The phase transition is likely gradual; our parameter sweep is too restricted to see intermediate behavior.
- In general, curvature makes global alignment more difficult.

Animations?

Embedding animations in a PDF is a pain. I've gathered some still preview images for completeness. Pivoting to live demonstration!



In Conclusion

- Successfully implemented a C++/Python pipeline for simulating interacting ABPs on arbitrary 2D Riemannian manifolds.
- Quantitatively demonstrated phase-transition across parameter sweeps for different manifolds.
- In general, it seems it is more difficult for particles to align on curved surfaces.
- Further testing with larger parameter sweeps, investigations into initial particle density, etc, will be necessary to codify this result.

Thank You

Questions?